Summary

In eukaryotes, reactions of aerobic energy occur in mitochondria. An inner membrane separates the mitochondria into two spaces: matrix and intermembrane space. ETC process in the innermembrane oxidises NADH and succinate, utilizing oxygen and generates ATP. The scheme of respiratory chain and its coupling to ATP production can be summarized as follows:

1. Chemical reactions that transfer electrons between reactants are called oxidation-reduction reactions, or redox reactions.

2. During oxidation, a substance loses electrons, or is oxidized. In reduction, a substance gains electrons, or is reduced.

3. Electron transport chain is a step wise process comprising of four complexes. A series of flavoproteins, cytochromes, iron-sulfur clusters and quinone constitute the electron transport chain.

4. Complex I transfers electron from NADH to the electron carrier ubiquinone and complex II transfers electrons from succinate to ubiquinone

5. Complex III is comprised of three important cytochromes known as cytochromes bL, bH and c plus an iron sulphur center. These transfer electrons from reduced ubiquinone (UQH2) to cytochrome c

6. Complex IV contain two cytochromes (a and a3), two copper atoms, and transfers electrons to oxygen

7. Several poisons such as MPTP, rotenone, azide and cyanide inhibit the electron transport chain

8. During oxidative phosphorylation electrons derived from NADH and FADH2 combine with oxygen and the energy released is used to drive the synthesis of ATP from ADP.

9. ATP synthase is a reversible coupling device that can convert the energy of the electrochemical proton gradient into chemical bond energy or vice versa.

10. The transfer of energy from $NADH/FADH_2$ to O_2 yields high energy of ΔG equal to -52.5kcal/mol from each pair of electrons transferred.

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